DSC 40A - Group Work Session 1

due Thursday, July 2nd at 11:59PM

Write your solutions to the following problems in LaTeX. One person from each group should submit your solutions to Gradescope. Tag all group members so everyone gets credit.

This worksheet won't be graded on correctness, but rather on good-faith effort. Even if you don't solve any of the problems, you should include some explanation of what you thought about and discussed, so that you can get credit for spending time on the assignment.

In order to receive full credit, you must work in a group of four students for at least 90 minutes in your assigned discussion section. You can also self-organize a group and meet outside of discussion section for 80 percent credit. You may not do the groupwork alone.

1 Minimizers and Maximizers

We've seen that machine learning problems must first be formulated as mathematical problems. Many of these mathematical problems turn out to be *optimization problems*: finding the value that minimizes or maximizes a function. For a function of one variable $f(x)$, a value x^* is said to be a **minimizer** of $f(x)$ if

$$f(x^*) \leq f(x) \quad \text{for all } x.$$

Similarly, x^* is said to be a **maximizer** of $f(x)$ if

$$f(x^*) \geq f(x) \quad \text{for all } x.$$

Notice that a function can have multiple minimizers or maximizers. For example, a constant function like $f(x) = 5$ is minimized at all values of x, and it's also maximized at all values of x.

Problem 1.

Should the blank below be filled in with the word *minimizer* or *maximizer* or *neither*? Prove your result.

If x^* is a minimizer of $f(x)$ then it's a ______________ of $g(x) = 5f(x) + 3$.

Problem 2.

Should the blank below be filled in with the word *minimizer* or *maximizer* or *neither*? Prove your result.

If x^* is a minimizer of $f(x)$ then it's a ______________ of $g(x) = (f(x))^2$.

2 Inequalities

Suppose we have an ordered list numbers, $y_1, \dots, y_n$ such that $y_1 \le y_2 \le \dots \le y_n$.

The **midpoint** of $y_1, \dots, y_n$ is thus $\frac{y_1+y_n}{2}$, or the average of the minimum and maximum.

As such, we may deduce that the midpoint is at most y_n and is at least y_1. We can prove this with a chain of inequalities.

The first step is to show that the midpoint is at most y_n. We have established that $y_1 \le y_n$. Thus, we use this to our advantage. Namely, we add y_n to both sides of the inequality, then divide by two:

$$y_1 \le y_n \implies y_1 + y_n \le 2y_n \implies \frac{y_1 + y_n}{2} \le y_n.$$

The lefthand side of our final inequality matches our midpoint exactly. Ergo, we have shown that the midpoint cannot exceed y_n.

Now it's your turn to put this into practice!

Problem 3.

Prove that the midpoint is $\ge y_1$.

Problem 4.

Let's take this a step further. Suppose $y_1, \dots, y_n$ are all positive numbers. The **geometric mean** of $y_1, \dots, y_n$ is defined to be:

$$(y_1 \cdot y_2 \cdots y_n)^{1/n}.$$

Prove that the geometric mean is less than or equal to y_n and greater than or equal to y_1.

3 Empirical Risk Minimization

In this problem, consider the loss function

$$L(y,h) = \begin{cases} 1, & |y-h| > 1 \\ |y-h|, & |y-h| \le 1 \end{cases}.$$

Problem 5.

1. Write $L(y,h)$ as an explicit piecewise function of h, with each case fully simplified. In other words, let the constraints for expression in the piecewise function be contingent on h.
2. Prove or disprove: if $\max_i y_i - \min_i y_i \le 1$, then for any h between $\min_i y_i$ and $\max_i y_i$,

$$R(h) = \frac{1}{n}\sum_{i=1}^{n} L(y_i,h) \quad \text{equals} \quad \frac{1}{n}\sum_{i=1}^{n} |y_i - h|.$$

3. Using part (b) (or otherwise), find a minimizer of R for the dataset $\{2,\ 2.25,\ 2.5,\ 2.75\}$. No computation of R itself should be necessary.